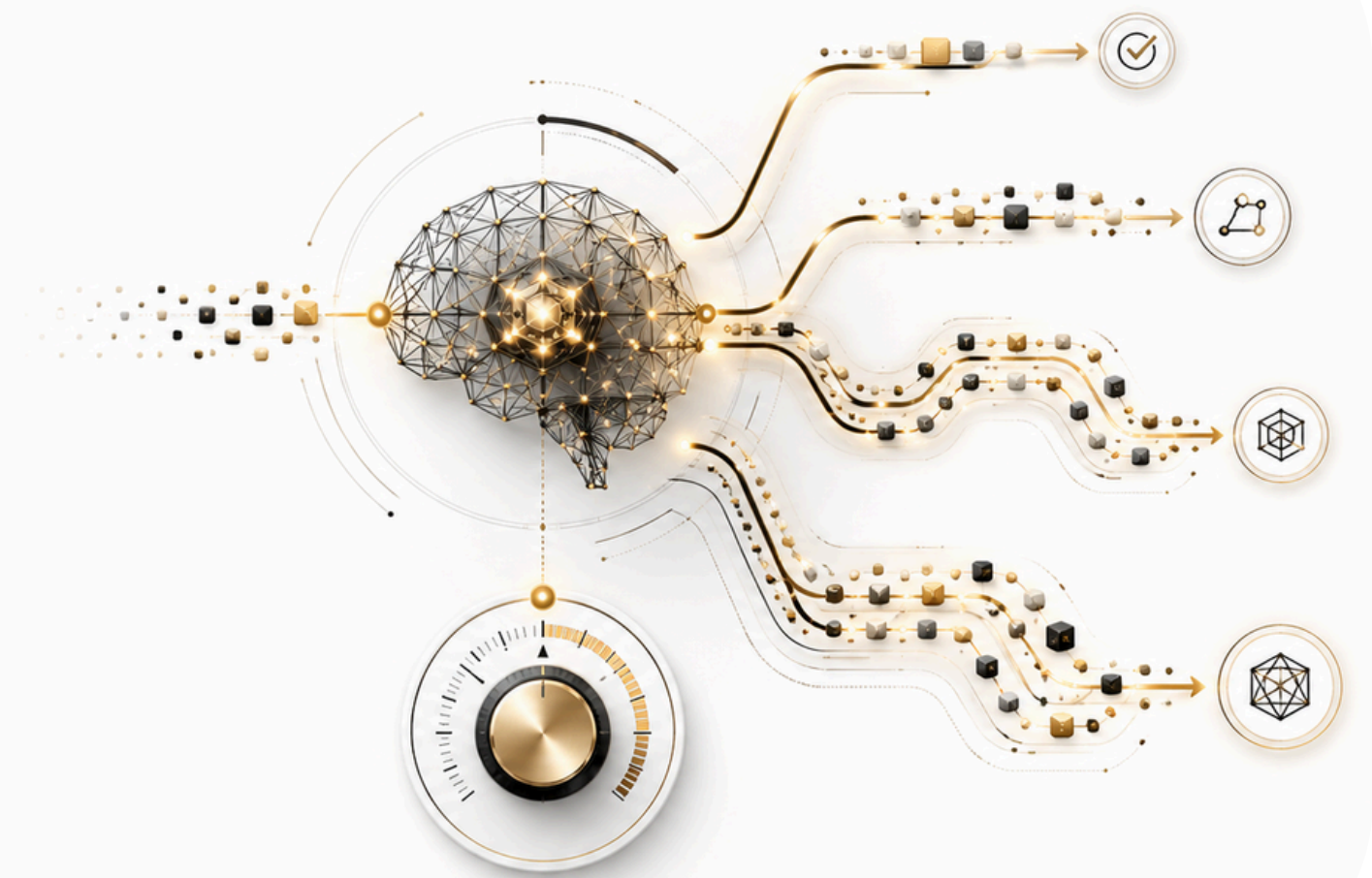


Adaptive Reasoning Budgeting for LLMs

Atharva Chaudhari



Introduction

LLMs need reasoning tokens to solve complex problems

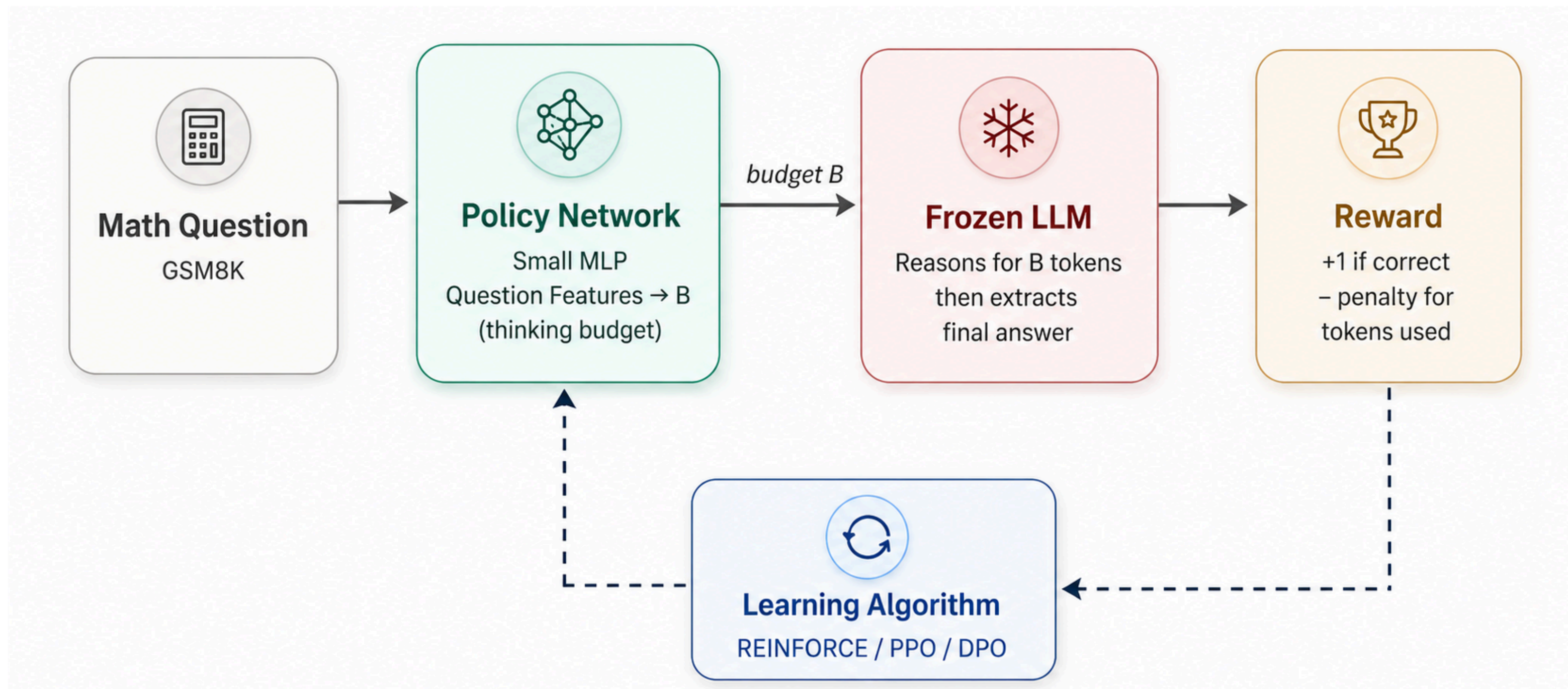
Fixed budgets give every question the same thinking time

Easy questions waste tokens

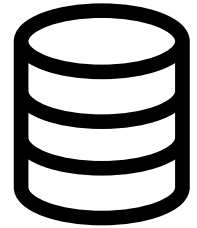
Hard questions may need more tokens

RL-based controller dynamically selects the reasoning-token budget

Methodology

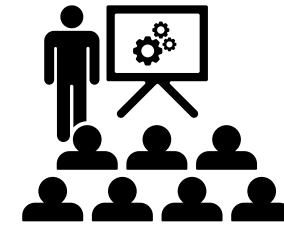


System Overview



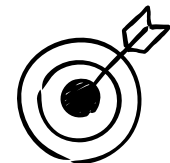
Dataset Setup

- GSM8K Math Problems
- Frozen Qwen2.5 Instruct LLM
- MLP Policy Network



Training

- Developed 10 Models
- M1-M4: RL Foundation
- M5-M9: PPO Based
- M10: DPO Based



Best Results

- M9 Achieved 80% Accuracy at 185 tokens
- Beat Fixed 224 token baseline using fewer tokens

Design Decisions

1

Cold Start Problem

Fix: Warm-Start

2

Wrong Signals

Fix: Minimum Correct
Budget

3

RL Damaged Warm-Start

Fix: High Budget
Exploration Floor

4

Slow LLM Inference

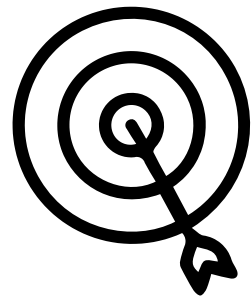
Fix: Batched GPU
Inference

5

PPO Instability

Fix: PPO Clipping +
Gradient Clipping

Model Development



Model 1 to Model 4

Learning Policy: Policy Gradient

Fixes: Warm Start + Reward

Best Result: M4 → 33% at 116 tokens, beat 128 Fixed Token



Model 5 to Model 9

Learning: PPO Based

Fixes: Better Token Budget

Best Result: M9 → 80% at 185 tokens, beat 224 Fixed Token

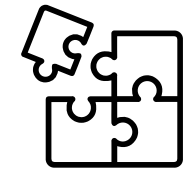


Model 10

Learning: DPO Based

Challenge: Preference Pair

Best 1.5B Model: M8



Model Setup

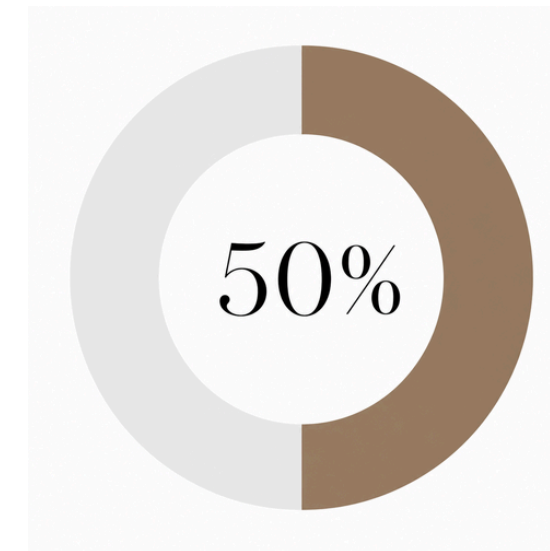
- PPO Policy with minimum correct budget
- Batched GPU Inference



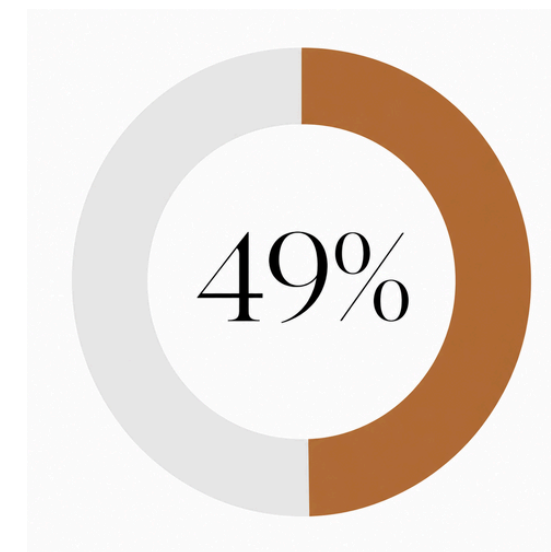
What it Achieve

- Budget Range from 213 to 241 Tokens
- Stable Difficulty aware allocation

Why Matters ?



Fixed 256 Tokens



Avg. 223 Tokens



Question:

What if we replace the 1.5B model with more powerful model?

Best Overall: 7B Model

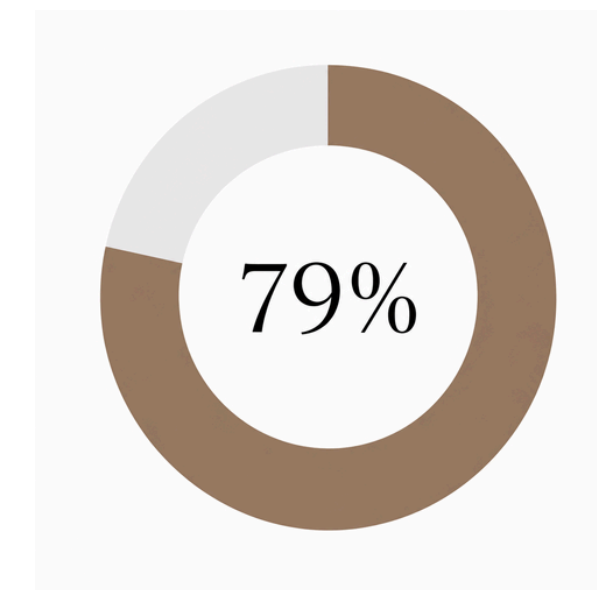
Model Setup

- Q-wen 7B model replaces 1.5B
- Batched GPU Inference

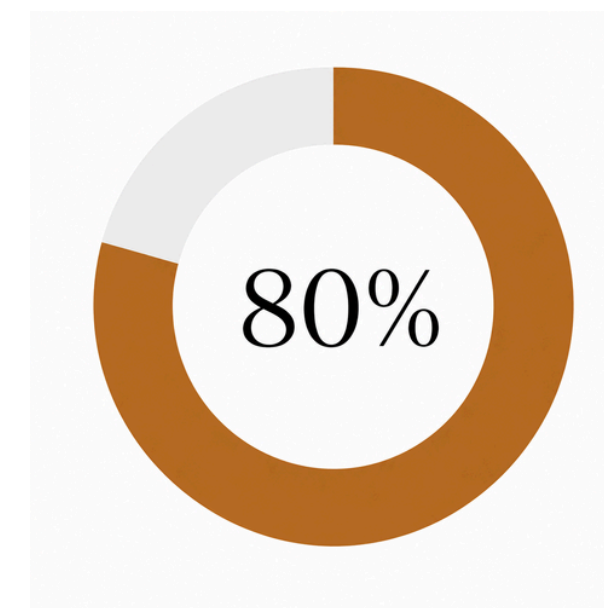
What it Acheive

- Budget Range from 175 to 192 Tokens
- Used approx 39 tokens less to beat the accuracy

Why Matters ?



Fixed 224 Tokens



Avg. 185 Tokens

Conclusion

Explored multiple methods: Reward Calculation, PPO, and DPO

Reward design mattered

DPO failed due to insufficient preference pairs when base model is too strong

Future work: larger datasets, stronger policy encoder, joint LLM and policy training

Thank You !!!!!